Homework

Homework 13 — Reservations and overbooking — 60 min.

A hall has eighteen seats. The organiser accepts twenty reservations, paid in advance. Each person who has reserved turns up with probability 0.8, independently of the others. Denote by X the number of people who turn up. If more than eighteen people come, the surplus people are turned away.

1. Justify the distribution of X, then calculate its expected value and variance.
2. Calculate the probability that the organiser has to turn away at least one person. Round to 10^{-6}.
3. Each reservation brings in 10 monetary units, even if the person does not come. Each person turned away receives a total payment of 30 monetary units, refund and compensation included. Denote by $R = \max(X - 18; 0)$ the number of people turned away. Express the profit G in terms of R, then calculate $E(G)$.
4. Calculate the profit if the organiser accepts only eighteen reservations, with the same payment rules. Compare the two strategies in terms of mean profit.
5. State one assumption of the model that may be questionable in practice, then a criterion other than monetary profit to take into account in this decision.